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FACULTY OF ENGINEERING
BILBAO

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ERREKA DSS PROJECT

Bilbao, 2026.

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Course: Sistema de Toma de Decisiones y Business Intelligence

ERREKA DSS PROJECT

Team 14

Lecturer:

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Summary:

The following project focuses on the development of a Decision Support System for Erreka, a global company that operates thousands of connected automatic doors across diverse environments including hospitals, industrial facilities, logistics centers, and residential complexes. The primary objective is to transform raw operational data into actionable business intelligence that enhances decision-making processes, identifies recurring technical issues, and improves overall service efficiency. The project moves beyond simple data analysis toward a structured, layered decision design grounded in real operational data.

Activity 1 established the conceptual foundation of the DSS by defining the core decision problem — how to prioritize incident response across a global installed base under limited service resources — and identifying the decision-maker, organizational level, decision criteria, and a four-layer conceptual architecture consisting of Decision, Information, Logic, and Interface layers.

Activity 2 focused on the technical infrastructure required to support the DSS. A twelve-table Entity-Relationship model was designed following a star schema principle, with `incident_events` as the central fact table surrounded by dimension and log tables. A reusable Python-based ETL pipeline was implemented to extract data from CSV sources, apply necessary transformations, and load all twelve datasets into a MySQL database while enforcing referential integrity and supporting repeated execution without data loss.

Activity 3 built a business reporting system on top of the data warehouse, producing a set of Grafana dashboards covering incident distribution by alert type, criticality level, SLA category, customer type, and country priority risk. These dashboards provided the first structured view of Erreka's operational reality, revealing that 787 out of 998 total incidents fall under Critical SLA contracts and that the vast majority of incidents originate from B2B clients.

Activity 4 moved from reporting to descriptive analytics by organizing the analysis around three core questions and an explicit drill-down path across four levels: Global, Segmentation, Context, and Product Mix. Key insights included the identification of a structural concentration risk in three markets — the United States, China, and India — that together disproportionately exceed the global benchmark of 3,435 doors per country, the discovery that B2B installations carry a 2:1 volume advantage over B2C with full SLA exposure, and the confirmation that each door type maps exclusively to one customer segment, making product line and customer segment analytically interchangeable dimensions.

Activity 5 introduced predictive analytics by implementing a rule-based risk classification model as a MySQL VIEW, assigning each door a risk score of High, Medium, or Low based on failure history, days since last failure, criticality level, and SLA category. The resulting dashboard identified 76 High risk doors globally, confirmed that ROLLfast Industrial 300 carries approximately 50% of the total failure burden, validated the model through a perfect correlation between criticality level and maximum risk score, and surfaced a set of silent threat doors with documented failure history that standard monitoring would overlook. The final panel translated all risk scores into a ranked maintenance priority list that a maintenance manager can act on immediately, completing the full path from raw data to operational decision support.

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ACTIVITY 1: DESIGNING A DSS FOR ERREKA

Decision Definition

Decision Statement

The Decision Support System is designed to support the following operational decision: “Given multiple alerts and incidents generated by connected doors across the global installed base, which incidents should be handled first and with what urgency in order to minimize operational disruption, safety risks, and contractual violations under limited-service resources”

This decision is not about solving the technical problem itself but about prioritizing the sequence of response actions when many potential incidents occur simultaneously.

Decision-Maker

The primary decision-maker is the Global Service Control / Operations Center responsible for monitoring and coordinating service interventions across Erreka’s connected infrastructure. This operational team continuously supervises alerts generated by thousands of automated doors deployed worldwide. Their role is to:

Monitor real-time operational alerts, Evaluate the seriousness of incidents, Decide which incidents require immediate response, Coordinate field technicians or remote interventions...

The decision is because of that human-centered but system-supported. The DSS does not replace operators; it structures their reasoning when multiple incidents compete for attention. Additional stakeholders may include some regional service managers, key customers, management teams responsible for SLAs...

Organizational Level

This decision belongs primarily to the tactical level of the organization because in my opinion they have all of the characteristics that we have been through in our classes which is mostly the fact that this decisions are human but with help of the automated systems. However, the consequences of these decisions may propagate upward into strategic levels, because repeated prioritization failures could lead to higher maintenance costs, SLA violations, loss of customers and so on.

Decision Frequency

The decision is taken continuously in real time.

Whenever multiple alerts occur simultaneously, the operations center must determine which alert should be addressed first and could response be delayed or not. Given the scale of Erreka’s installed base, this decision may be taken hundreds or thousands of times per day. Because of this high frequency, relying purely on human intuition becomes increasingly difficult.

Operational Context

Erreka manages a large network of **connected automated doors** deployed in diverse environments such as hospitals, retail stores, residential buildings...

Each environment presents very different operational risks.

For example a malfunctioning door in a hospital emergency entrance may have serious safety implications.

At the same time, Erreka’s service resources are limited. The company cannot respond to all incidents simultaneously because of travel time, lack of technicians...

The decision environment is therefore characterized by multiple simultaneous alerts, limited operational resources, time pressure and so on.

These characteristics make the decision semi-structured, meaning that human judgment is necessary but can be supported by structured information and reasoning.

Consequences of a Wrong Decision

Incorrect prioritization of incidents may produce significant negative consequences such as:

1. Safety risks, for example:

- Emergency exits that cannot open

- Security doors that remain unlocked

- Industrial doors that malfunction during operations

Failing to prioritize these incidents may lead to accidents or legal liability.

Operational disruption because If low-impact incidents are handled before high-impact ones, the result can be unnecessary operational disruption for clients.

2. SLA violations

Many service contracts include Service Level Agreements that define maximum response times.

Failure to respond within the agreed timeframe may result in financial penalties, contractual disputes, loss of customer trust...

3. Inefficient use of service resources

If technicians are dispatched to lower priority incidents while more critical ones remain unresolved, the company may experience bigger service costs, less efficiency...

4. Reputational damage

Repeated service failures can damage Erreka's reputation, particularly in critical sectors such as healthcare or logistics.

Therefore, the quality of incident prioritization directly affects customer confidence and long-term business relationships.

Objectives and Constraints

Primary Objective

The primary objective of the DSS is ensure that incident response actions are prioritized in a way that minimizes operational risk and service disruption across the installed base. This objective focuses on effective allocation of limited response capacity.

Secondary Objectives

In addition to the primary objective, several secondary objectives guide the decision process:

- 1. Improve response efficiency**

The DSS should help reduce the time required for operators to evaluate incidents by providing structured information and prioritization guidance.

This improves the efficiency of the operations center.

- 2. Optimize the use of service resources**

Field technicians, travel capacity, and technical expertise are limited resources.

The DSS should help ensure that these resources are allocated where they generate the **highest operational value**.

- 3. Reduce cognitive overload for operators**

Monitoring thousands of connected devices can generate a large volume of alerts.

Without structured support, operators may experience decision fatigue, increased response times, and a higher likelihood of overlooking critical incidents amid a flood of lower-priority alerts. This cognitive burden directly reduces the quality and consistency of prioritization decisions, particularly during peak alert periods.

- 4. Support consistent decision-making**

Different operators may interpret the same situation differently.

The DSS should promote **consistent decision logic**, ensuring that similar incidents are treated similarly across shifts and regions.

Organizational Constraints

The DSS must operate within several organizational constraints, Erreka already has established procedures for incident reporting or technician dispatch. The DSS must align with these processes rather than replacing them.

Operational Constraints

Limited technician availability

At any given time, the number of technicians available for interventions is limited.

This constraint makes prioritization essential.

Geographic dispersion

Doors are installed across multiple countries and regions.

Travel time significantly influences how quickly an incident can be resolved.

Technical feasibility of remote resolution

Some incidents can be resolved remotely through software resets or configuration changes.

Others require **physical intervention on site**.

The DSS must consider the feasibility of different response actions.

Economic Constraints

Service costs

Dispatching technicians involves labor costs, travel expenses, and of course the fact that technician is unavailable while he is on the task

Contractual obligations

Different clients have different service agreements.

Premium contracts may require faster response times, which influences prioritization decisions.

Decision Criteria

To support consistent prioritization, the DSS evaluates incidents according to several conceptual criteria. These criteria represent dimensions of comparison, not numerical metrics.

Safety and Security Risk

This criteria evaluates whether the incident poses a risk to people, property, or security. Such as emergency doors that can't open, security doors that remain unlocked... It is important to say that the incidents involving safety risks must receive immediate attention, often overriding other criteria.

Operational Impact

This criteria assesses how much the door failure affects the client's operations.

The same technical malfunction may have very different consequences depending on context.

Understanding the business context of the door is therefore essential.

SLA Exposure

This criteria evaluates the contractual obligations associated with the installation.

Certain customers have strict service agreements that require rapid response.

If the response time approaches the SLA limit, the incident becomes more urgent in order to avoid financial penalties or customer dissatisfaction.

Incident Persistence

Some alerts disappear quickly or resolve automatically.

Others persist over time and indicate unresolved technical problems.

Persistent incidents may require higher priority because they indicate system instability or unresolved failures.

Response Feasibility

Some incidents can be resolved quickly through remote actions, while others require complex on-site repairs.

This criterion evaluates whether an intervention is quickly achievable with available resources.

In certain situations, resolving several smaller incidents quickly may free resources to address more complex problems later.

Resource Availability

The availability of technicians, tools, and spare parts influences the feasibility of response.

Even high-impact incidents may experience delays if required resources are unavailable.

This criterion ensures that prioritization decisions remain operationally realistic

Conceptual DSS Architecture

The proposed DSS follows a layered conceptual architecture aligned with classical Decision Support System design.

The architecture includes four main layers:

Decision Layer

The decision layer defines the core purpose of the DSS. This layer was detaily described before in the document.

Information Layer

The information layer provides the contextual data necessary to support prioritization decisions.

Relevant categories of information include:

Real-time IoT telemetry - connected doors continuously generate operational data such as code of error, status changes, system alerts...

Asset registry

Information describing each installation, including location door type, usage...

Customer and contract information (SLA)

Historical incident data

Past records of incidents provide insight into maintenance history, known technical issues from past, this information helps technicians to put failure into context

Logic Layer

The logic layer transforms raw information into structured decision support.

At this conceptual stage, the logic layer includes qualitative reasoning mechanisms rather than mathematical models.

Possible components include:

Rule-based prioritization logic - Safety-related alerts immediately escalate to highest priority or SLA-critical incidents receive increased urgency.

Contextual evaluation

The system combines multiple contextual factors to assess the seriousness of incidents like location type, customer hierarchy, operational environment...

Recommendation support

The DSS may suggest potential response actions, such as dispatch a technician or continue monitoring these suggestions assist the operator but do not replace final judgment.

Interface Layer

The interface layer defines how the decision-maker interacts with the system.

The goal of the interface is not to display data but to **support decision reasoning**.

Key functions include:

Structured incident overview

Operators receive a structured overview of current incidents, organized by priority rather than by arrival time.

This helps reduce cognitive overload.

Incident comparison support

The interface allows operators to understand why one incident is considered more critical than another.

This transparency increases trust in the system.

Contextual explanations

The system should provide explanations indicating which factors influenced prioritization.

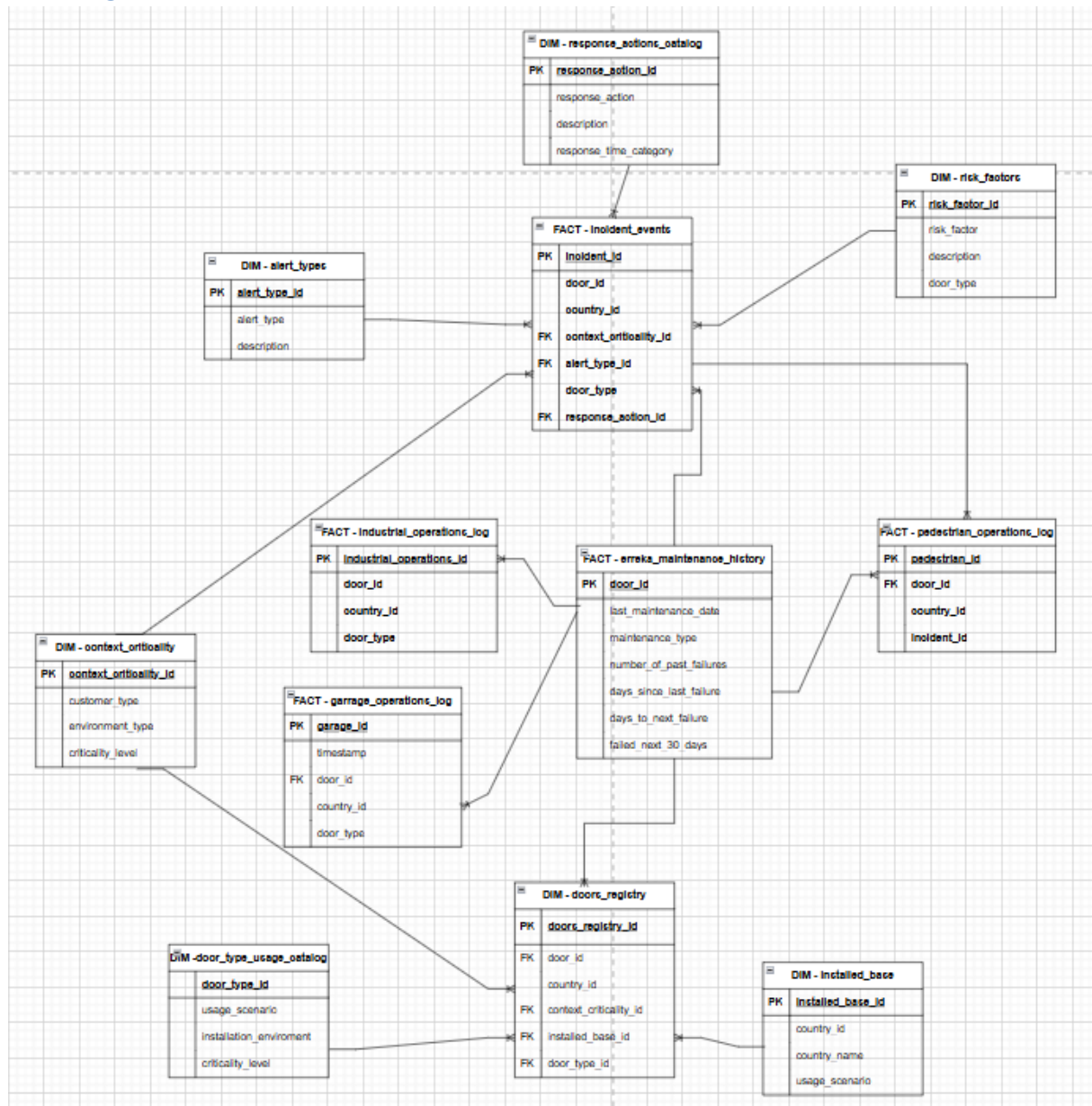
This supports human understanding and accountability.

Operator control

The operator retains full authority to override system recommendations when necessary.

Human expertise remains essential in complex or ambiguous situations.

ACTIVITY 2: TO DESIGN, IMPLEMENT, AND DOCUMENT A REUSABLE DESCRIPTIVE ANALYTICS PIPELINE FOR ERREKA DSS.



both `alert_types` and `context_criticality` through foreign keys, allowing each incident to be traced back to its alert category and criticality level. All three operational log tables — `garage_operations_log`, `pedestrian_operations_log`, and `industrial_operations_log` — contain `door_id` as an attribute, enabling each log entry to be traced back to a specific door, its type, and its country or usage scenario.

The tables `risk_factors`, `response_actions_catalog`, and `installed_base` are reference and catalog tables that are not directly linked through foreign keys at this stage. This is a deliberate design decision — these tables will be logically integrated into the DSS logic layer in later phases of the project.

The central fact table in the model is `incident_events`. This design decision is based on the nature of the data — each row in `incident_events` represents a single measurable event: a door incident that occurred at a specific time, in a specific context, with a specific alert type. All other tables either describe the context of that event (dimension tables such as `alert_types`, `context_criticality`, `doors_registry`) or record operational activity linked to doors over time (log tables such as `garage_operations_log` and `pedestrian_operations_log`). This structure follows the star schema principle, where a central fact table holds foreign keys to surrounding dimension tables, enabling efficient analytical queries across multiple dimensions such as criticality level, alert type, country, or door type. The choice of `incident_events` as the fact table is also analytically justified — incident prioritization is the core decision problem of this DSS, meaning that all meaningful analysis ultimately revolves around incidents and their context.

ETL process design

The ETL pipeline for the Erreka DSS is designed as a reusable Python-based process that can be executed multiple times without manual intervention, making it suitable for future tasks.

Extract Data is extracted from twelve CSV files stored in a configurable local folder. The source path is defined as a single variable at the top of the script, meaning that for future datasets, only that path needs to be updated. Each CSV file corresponds directly to one table in the Data Warehouse.

Transform Before loading, the following transformations are applied. Timestamp and date columns are parsed into proper MySQL-compatible formats. Empty or missing numeric fields are converted to NULL rather than left as empty strings. Surrogate primary keys that do not exist in the source CSVs (such as `maintenance_id`, `garage_id`, `pedestrian_id`) are generated by MySQL using `AUTO_INCREMENT`, so they are not manually assigned in the script. No aggregation is performed at this stage, as the goal is to preserve raw data fidelity in the warehouse.

Load Each transformed dataset is loaded into its corresponding MySQL table using `INSERT IGNORE`, which prevents duplicate rows on repeated executions. Tables are loaded in dependency order: dimension tables first (such as `alert_types` and `context_criticality`), followed by the core `doors_registry` table, and finally fact and log tables that reference it.

Foreign key constraints are enforced at the database level to ensure referential integrity.

Reusability The script is fully parameterized through a configuration section at the top of the file. To process a new simulation dataset, the analyst only needs to place the new CSV files in the configured folder and re-run the script. All logic is encapsulated in dedicated functions per table, making it straightforward to add new tables in future course activities.

Implementation and Documentation of the ETL

The ETL process was implemented as a single reusable Python script (`erreka_activity2.py`). The script is structured into four clearly separated functions: `create_tables()`, `extract_all()`, `load_all()`, and `validate()`, each responsible for one phase of the pipeline.

The `create_tables()` function creates all twelve tables using `CREATE TABLE IF NOT EXISTS`, meaning the script can be executed multiple times without destroying existing data (That part of the script was executed because I had some tables in my workbench from previous tasks and activities). The `extract_all()` function reads each CSV file using the pandas library, applying necessary transformations such as timestamp parsing, date formatting, and numeric type casting for missing values. The `load_all()` function inserts data into MySQL in dependency order — dimension tables first, followed by the core `doors_registry` table, and finally fact and log tables. The `validate()` function runs row count checks across all tables and executes a JOIN query to confirm that relationships between tables are working correctly. The full script is submitted as a separate file accompanying this document.

Populate the DB with the datasets

The ETL pipeline was executed using the implemented Python script against the `erreka_dss` MySQL database. Prior to loading, the script automatically verified that all CSV column names matched the corresponding table headers and that data types were consistently transformed — timestamps were parsed to DATETIME format, date fields converted to DATE, and numeric columns cleaned of empty string values.

All twelve datasets were successfully loaded into their corresponding tables, respecting primary key constraints and foreign key relationships. The loading order was enforced programmatically to ensure referential integrity — dimension tables were populated before fact and log tables. Errors such as duplicate primary keys were handled explicitly and logged to the console rather than silently ignored.

After loading, row counts were verified for each table and SELECT queries with JOIN operations were executed to confirm that relationships between tables were functioning correctly. The full validation output and query results are included as screenshots in the evidence section below.

EVIDENCE AND SCREENSHOTS*

SQL File 27* SQL File 28* SQL File 29*

Limit to 1000 rows

```

1 • SELECT 'installed_base' AS table_name, COUNT(*) AS row_count FROM installed_base
2 UNION ALL
3 SELECT 'door_type_usage_catalog' AS table_name, COUNT(*) AS row_count FROM door_type_usage_catalog
4 UNION ALL
5 SELECT 'risk_factors' AS table_name, COUNT(*) AS row_count FROM risk_factors
6 UNION ALL
7 SELECT 'alert_types' AS table_name, COUNT(*) AS row_count FROM alert_types
8 UNION ALL
9 SELECT 'context_criticality' AS table_name, COUNT(*) AS row_count FROM context_criticality
10 UNION ALL

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

table_name	row_count
installed_base	120
door_type_usage_catalog	18
risk_factors	15
alert_types	10
context_criticality	6
response_actions_catalog	10
doors_registry	504
pedestrian_operations_log	895268
industrial_operations_log	1004548
garage_operations_log	43981
incident_events	998
erreka maintenance historv	504

Result Grid
Form Editor
Field Types

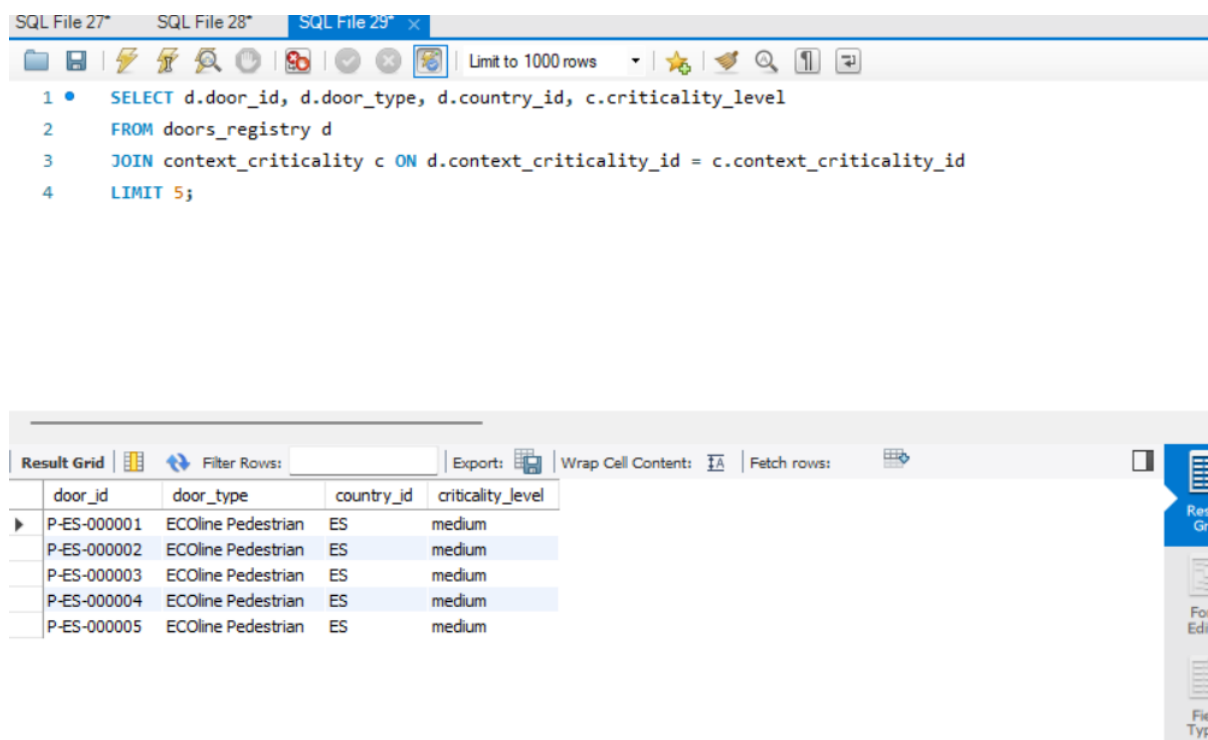
```

1 • SELECT i.incident_id, i.door_id, i.alert_type_id, c.criticality_level, c.sla_category
2 FROM incident_events i
3 JOIN context_criticality c ON i.context_criticality_id = c.context_criticality_id
4 LIMIT 5;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: |

incident_id	door_id	alert_type_id	criticality_level	sla_category
INC-P-ES-000003-155	P-ES-000003	AT01	medium	Standard
INC-P-ES-000003-283	P-ES-000003	AT01	medium	Standard
INC-P-ES-000004-202	P-ES-000004	AT03	medium	Standard
INC-P-ES-000005-81	P-ES-000005	AT02	medium	Standard
INC-P-ES-000007-199	P-ES-000007	AT02	very high	Critical



MySQL Workbench 2026-03-16 11-48-00.mp4



- erreka_activity2.py - Decision support systems - Visual Studio Code 2026-03-16 11-58-51.mp4

*I didn't show you full process of running a script in python because loading of data into tables lasted 3-4 minutes when I did it the first time so I didn't want you to waste time waiting just to script to load but here is the image of final output:

```

alert_types          10
context_criticality  6
door_type_usage_catalog 18
response_actions_catalog 10
risk_factors         15 ...

garage_operations_log 43981
pedestrian_operations_log 895268
garage_operations_log 43981
pedestrian_operations_log 895268
industrial_operations_log 1004548

Spot check – incident_events JOIN context_criticality (3 rows):
pedestrian_operations_log 895268
industrial_operations_log 1004548

Spot check – incident_events JOIN context_criticality (3 rows):
industrial_operations_log 1004548

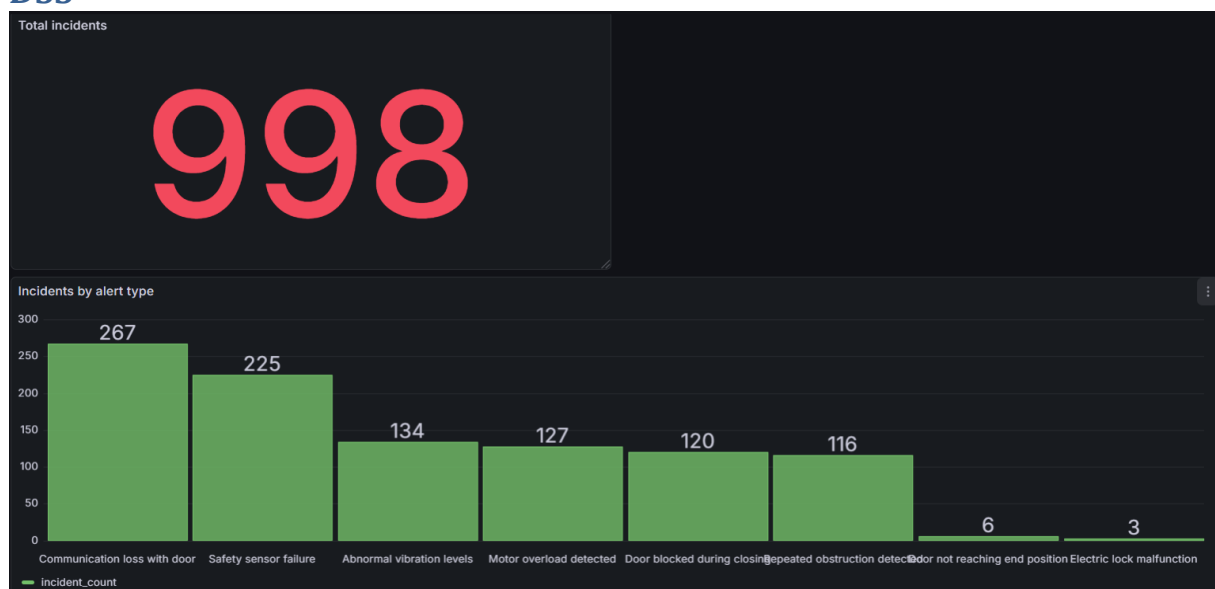
Spot check – incident_events JOIN context_criticality (3 rows):

Spot check – incident_events JOIN context_criticality (3 rows):
('INC-P-ES-000003-155', 'P-ES-000003', 'medium')
('INC-P-ES-000003-283', 'P-ES-000003', 'medium')
('INC-P-ES-000004-202', 'P-ES-000004', 'medium')
('INC-P-ES-000004-202', 'P-ES-000004', 'medium')

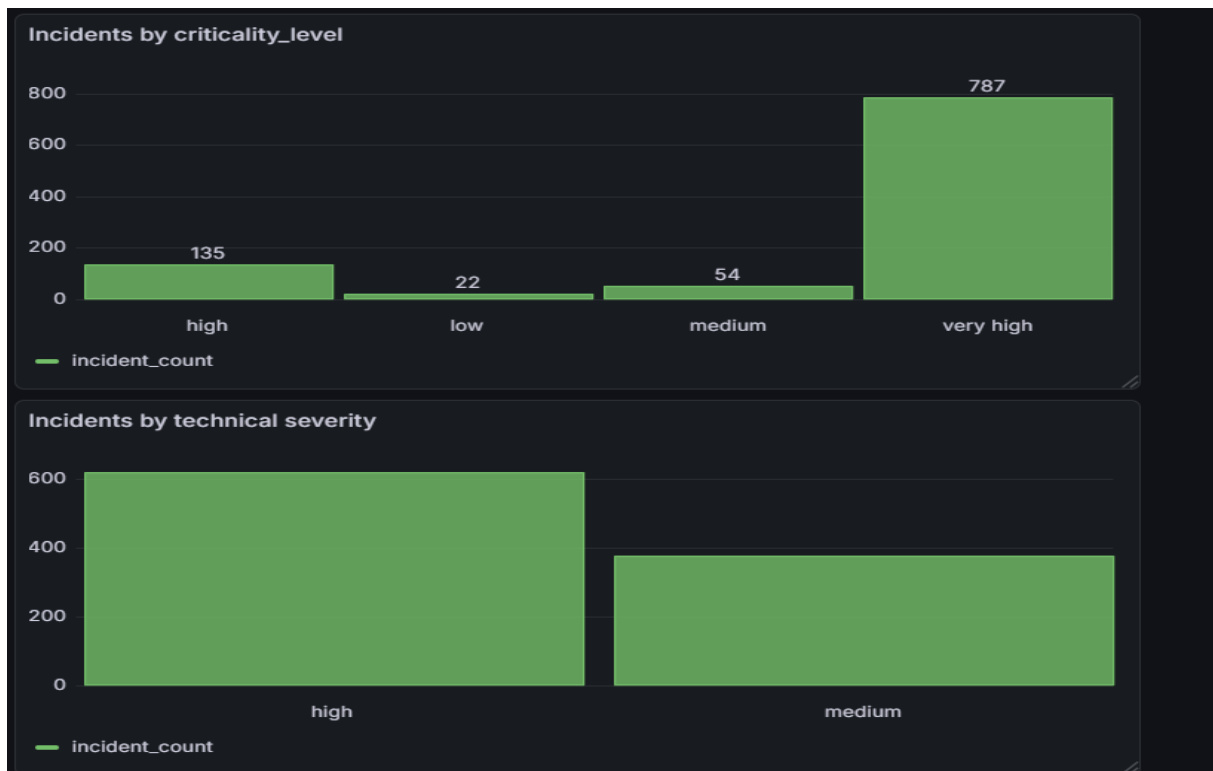
✓ ETL pipeline completed successfully

```

Activity 3: BUILDING A BUSINESS REPORTING SYSTEM FOR ERREKA DSS

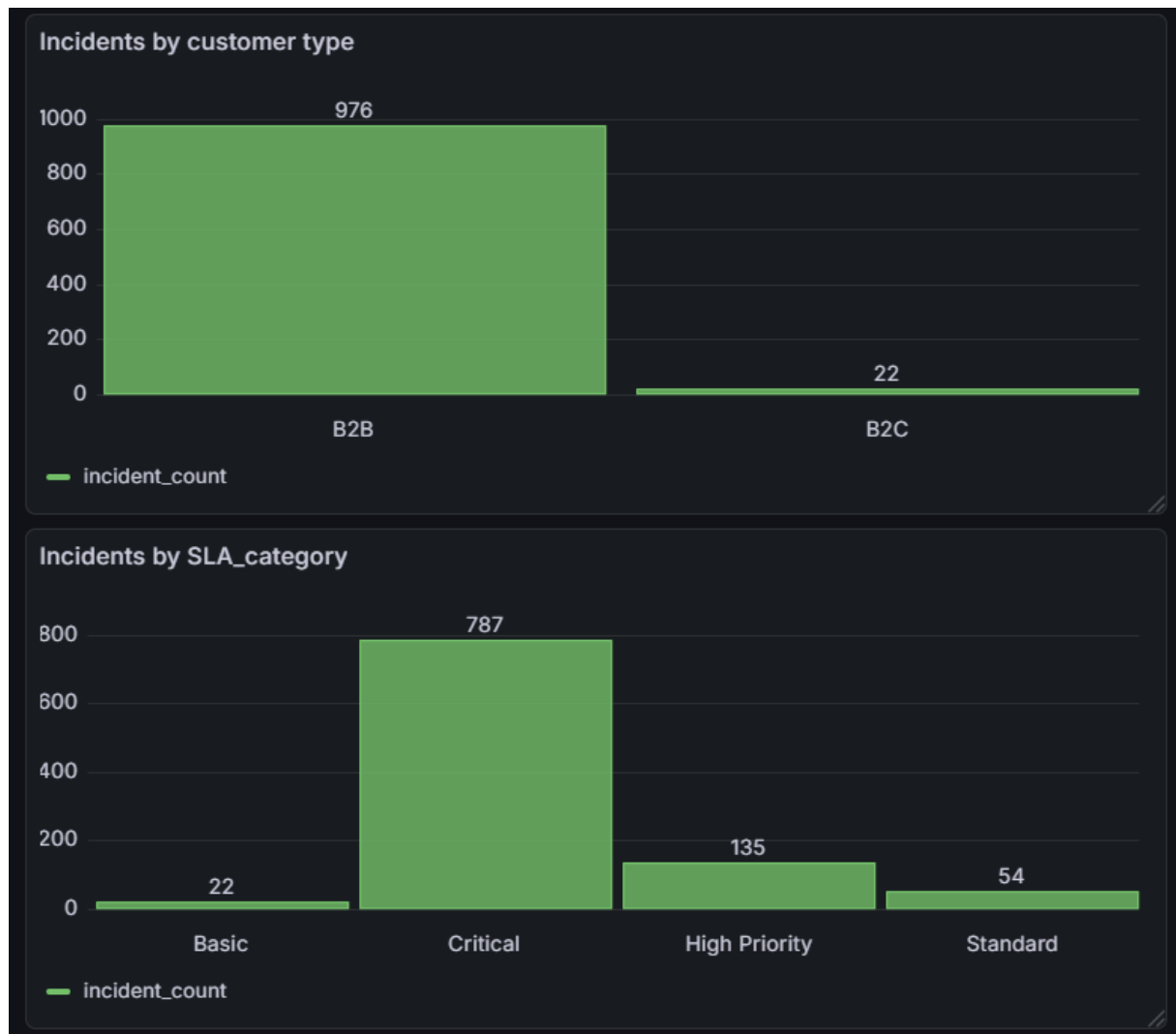


At the top you can see the total number of incidents (998) for the system, the bar chart below shows how many incidents there are for each alert type: most of them are due to communication loss with the door and safety sensor failures, while other types (abnormal vibration, motor overload, door blocked during closing, etc.) occur less often.



The first (top) chart shows the distribution of incidents by criticality level: most incidents fall into the very high category, followed by high, while medium and especially low criticality incidents are much less frequent.

The second chart shows incidents by technical severity: incidents are split between medium and high severity, with high-severity incidents forming the larger share of the total.



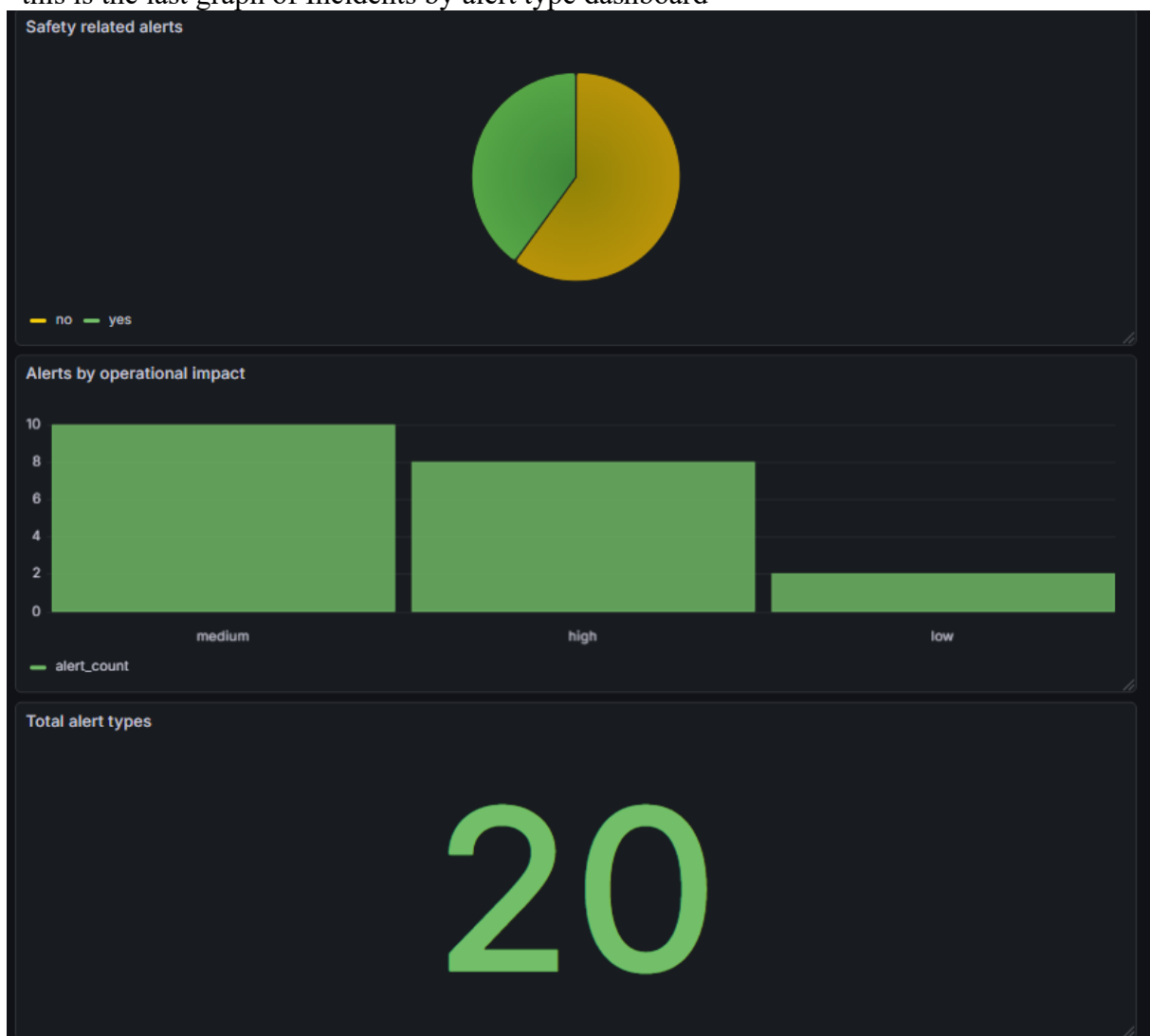
The first chart shows incidents by customer type. Most incidents come from B2B clients (976), while only a few are from B2C customers (22). This means the main operational focus is on business clients, where most service issues occur.

The second chart shows incidents by SLA category. Most incidents (787) are under Critical SLAs, meaning they require the fastest response to avoid penalties. Fewer cases belong to High Priority (135), Standard (54), and Basic (22). This helps the operations team identify which incidents need the most urgent attention.



This chart shows the number of incidents over time. It illustrates short-term fluctuations in alerts, with a drop on the second day followed by an increase on the third. This helps identify periods of higher activity and can guide when more monitoring or technical staff might be needed.

*this is the last graph of Incidents by alert type dashboard



The first chart shows the proportion of safety-related alerts compared to non-safety alerts. Around one-third of alerts are linked to safety issues, helping operators quickly identify incidents that may pose immediate risks.

The second chart displays alerts by operational impact. Most alerts are classified as medium or high impact, meaning they can cause noticeable disruption to clients if not handled quickly. The third panel shows the total number of alert types in the system, which is 20. This provides an overview of the variety of alert categories monitored by the DSS.

*This are the all screenshots of Alert Types Analysis dashboard



The first chart shows the distribution of contexts by SLA category. Most contexts are classified as Standard or Critical, meaning they are the main service levels that define how quickly incidents must be resolved.

The second chart shows contexts by criticality level. Most are medium or very high, which indicates that many doors operate in environments where failures could have significant impact.

The last panel shows the total number of defined contexts, which is 6. This gives an overview of how many unique operating contexts the DSS uses for decision support.

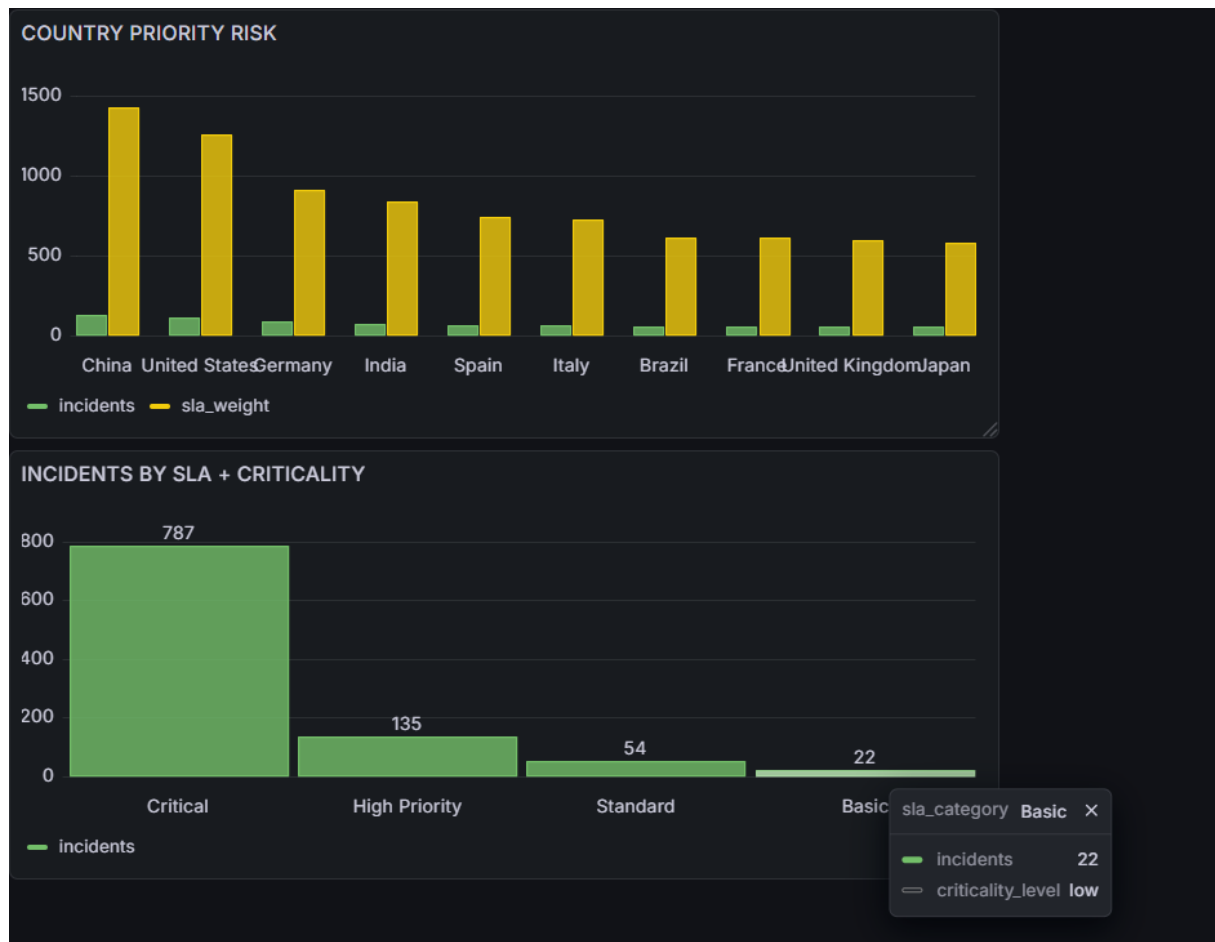
*This are the all screenshots of Context Criticality Analysis dashboard



The first chart shows the distribution of doors by type. The three main types—ROLLfast Industrial 300, ECOLine Home, and ECOLine Pedestrian—are evenly represented, giving a clear overview of the product mix across installations.

The second chart shows doors by country. The largest numbers of installations are in the United States, China, and India, followed by several European countries. This helps understand where most connected doors are located and where incidents are more likely to occur.

The last panel shows the total number of installed doors, which is 1008. This represents the full installed base managed by the DSS.



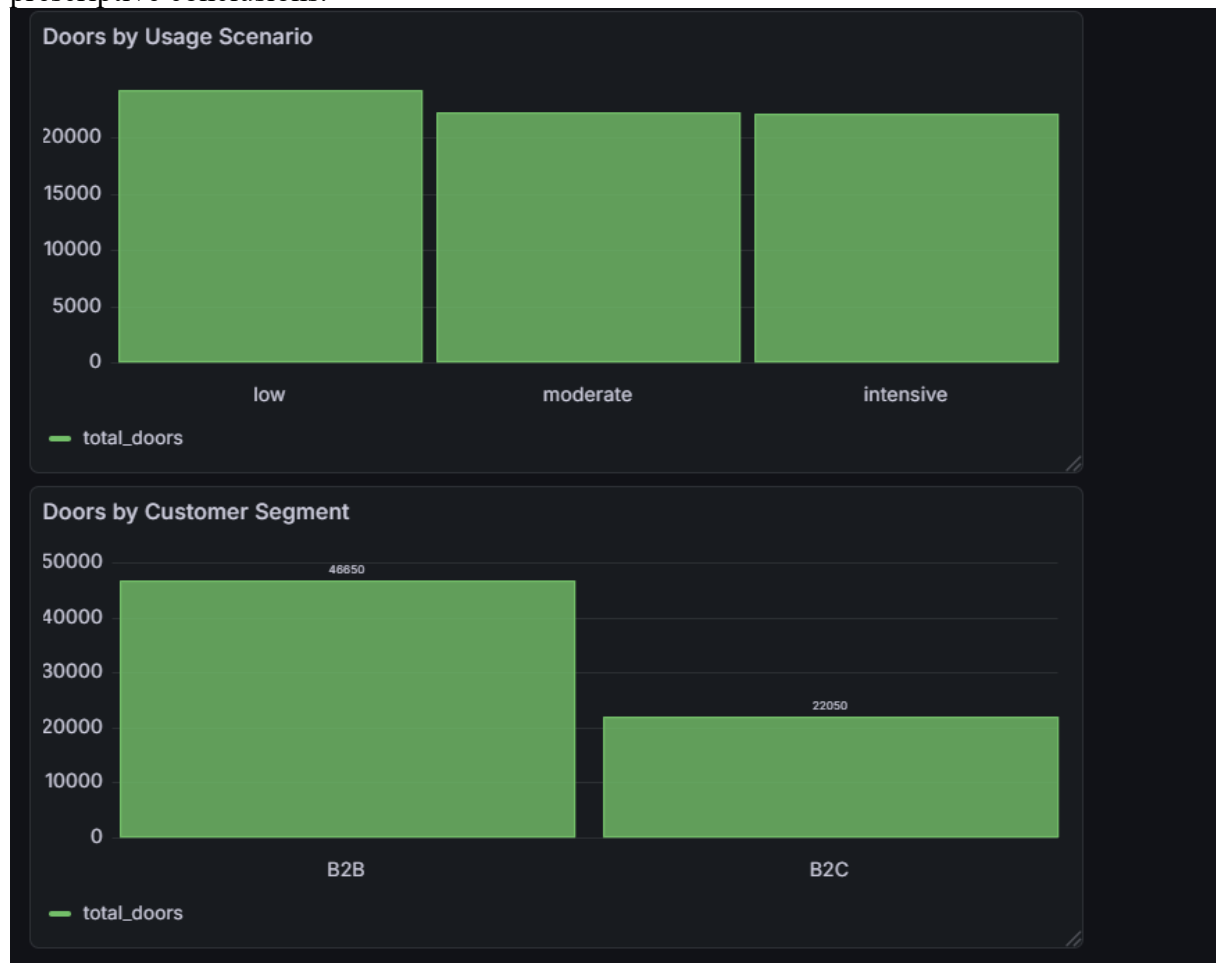
The 'Country Priority Risk' and 'Incidents by SLA and Criticality' charts shift the reporting focus from general volume to operational urgency. The 'Country Priority Risk' chart identifies specific markets where high-criticality installations are concentrated, allowing for strategic resource allocation—for instance, ensuring that technical teams in high-risk countries are equipped with specialized spare parts for the most sensitive environments. Complementing this, the 'Incidents by SLA and Criticality' chart acts as a tactical decision-making tool; by isolating 'Very High' criticality incidents within 'Critical' or 'Emergency' SLA categories, the system pinpoint-identifies the 'failure-to-comply' risks that require immediate intervention. Instead of treating all incidents equally, this integrated view dictates a clear hierarchy of action: technical response must be prioritized for the segments where operational impact and contractual obligations overlap at their highest levels, thereby minimizing both safety risks and financial penalties



AVCITY 3 ERREKA VIDEO.mp4

ACTIVITY 4: FROM DATA TO DESCRIPTIVE INSIGHTS

This dashboard is structured around three core analytical questions: Where is Erreka's installed base concentrated, and does that distribution create operational risk? Which usage and customer segments generate the highest service burden? And which combinations of installation environment and door type require disproportionate operational attention? To answer these questions systematically, the dashboard follows an explicit drill-down path across four levels: Global, segmentation, context and product mix. A user navigating this dashboard begins at the global level to identify concentration patterns, then moves through successive levels to identify which specific segments and product combinations drive those patterns — progressively narrowing from a macro overview to a root-cause explanation. Each panel is designed to answer one specific analytical question, and insights are presented before recommendations to ensure that descriptive evidence is fully established prior to any prescriptive conclusions.



Doors by Usage Scenario

This panel answers the question: "Is the installed base evenly distributed across usage intensity levels, and does that distribution create an uneven maintenance burden?"

Segment - low vs. moderate vs. intensive usage. Baseline: total ~68,700 doors. All three categories are nearly equally represented — low a bit more than moderate and intensive doors. However, equal distribution does not mean equal operational burden: intensive-use doors generate significantly more daily open/close cycles than low-use doors, directly increasing failure probability. The key issue is that equal resource allocation across all three segments means intensive installations currently receive the same maintenance budget as low-use doors, despite carrying a substantially higher mechanical load.

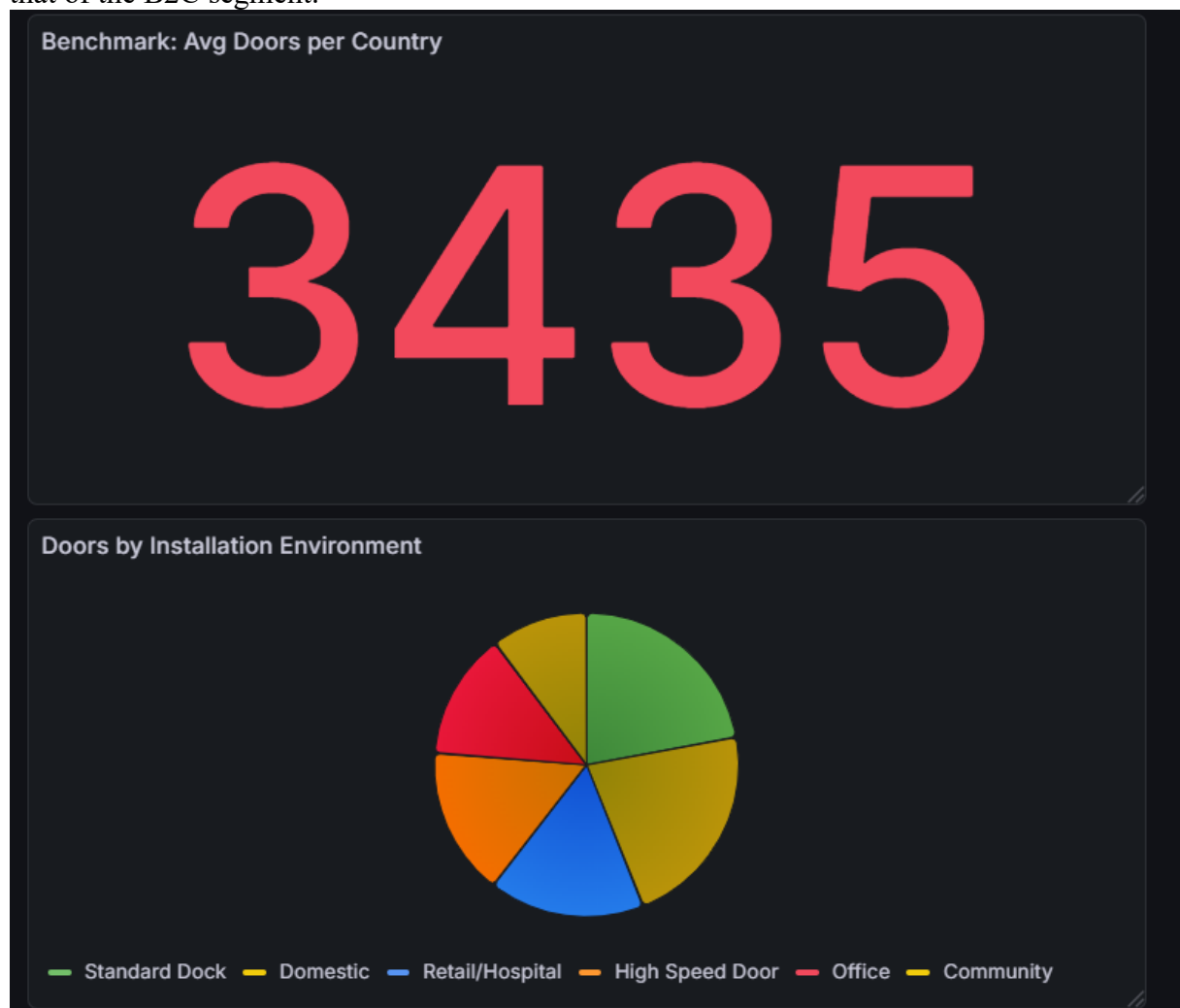
Recommendation - Erreka should adopt usage-segmented maintenance strategies, applying proactive inspection schedules to the intensive segment while allowing the low segment to operate on a reactive basis.

Doors by Customer Segment

This panel answers the question: "How is the installed base split between B2B and B2C clients, and what does that mean for operational risk exposure?"

Segment: B2B vs. B2C. Baseline: total 68,700 doors. B2B holds 68% versus B2C with 32% - a 2:1 ratio. This gap is operationally significant: every incident in the B2B segment carries formal SLA obligations with defined response times and financial penalties for non-compliance, while B2C incidents carry no such contractual exposure in general. With 46,650 doors under formal contracts, even a small increase in B2B incident response times translates directly into financial and reputational risk.

Recommendation: Erreka should configure all primary incident response resources around B2B needs, as the financial exposure from SLA breaches in this segment is more than twice that of the B2C segment.



Benchmark: Avg Doors per Country

This panel answers the question: "What is the global average installation density per country, and how can it serve as a reference point to identify over- and under-served markets?"

Segment - individual countries vs. global average. Baseline: 20 countries in the installed base, total around 70,000 doors. The global average of 3,435 doors per country immediately reveals a concentration problem: the United States (8,500), China (8,000), and India (6,000) all significantly exceed this benchmark, while countries like Portugal, Chile, Morocco, and Egypt

sit well below it. This means that 3 out of 20 countries account for a disproportionate share of the total installed base, creating a structural dependency risk where any service disruption in one of those three markets has an outsized impact on overall operational performance.

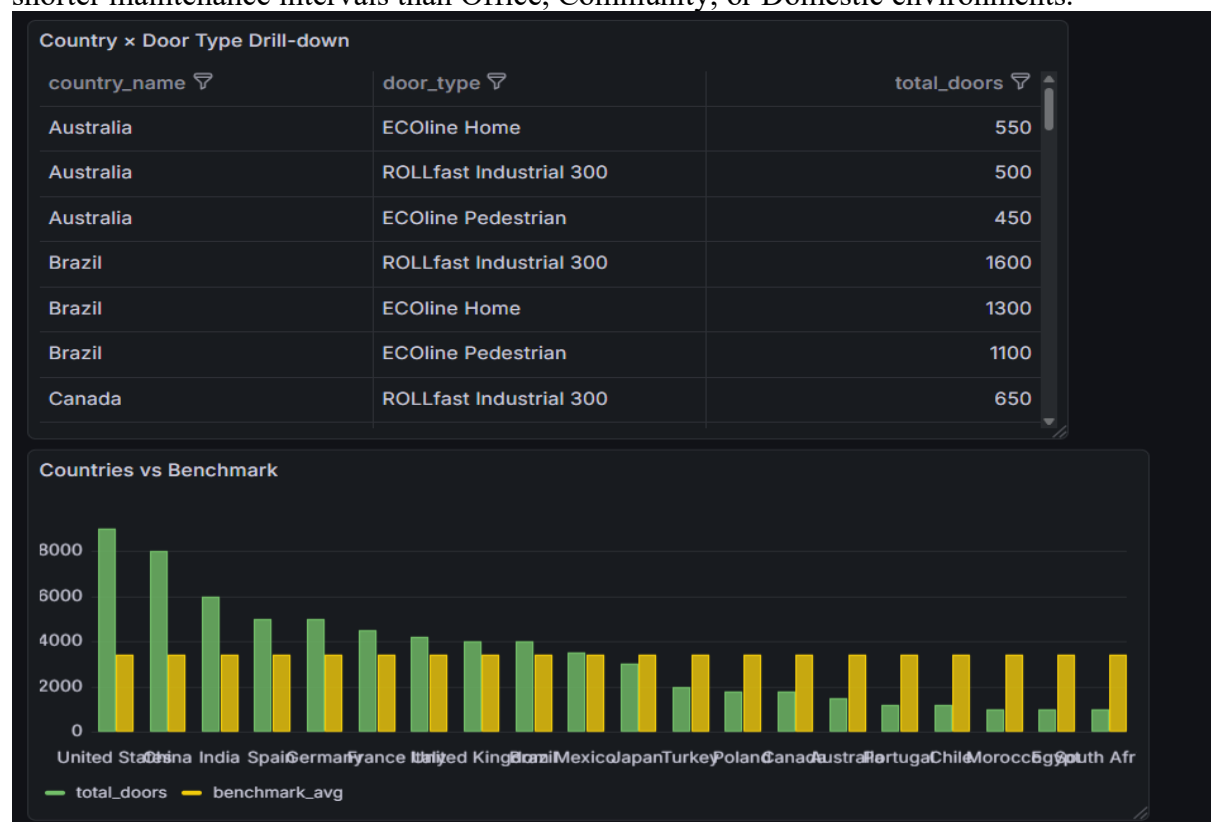
Recommendation - Erreka should use this benchmark actively in resource planning — countries above 3,435 require proportionally higher maintenance coverage, while countries significantly below it represent either untapped growth opportunities or markets requiring targeted investment.

Doors by Installation Environment

This panel answers the question: *"Are all installation environments operationally equivalent, or do certain environments demand disproportionately higher service attention despite similar installation volumes?"*

Segment - six installation environments (Standard Dock, Domestic, Retail/Hospital, High Speed Door, Office, Community). Baseline: total ~68,700 doors distributed relatively evenly across all six categories. The balanced distribution across environments should not be interpreted as operationally equivalent: Retail/Hospital and High Speed Door environments carry significantly higher safety and availability requirements than Domestic or Community installations. Concretely, a door failure in a Retail/Hospital environment creates both a safety risk and a potential SLA breach simultaneously, while the same failure in a Domestic environment carries neither of those consequences. Despite representing a similar share of the installed base, these two environment types demand fundamentally different response speeds and maintenance frequencies.

Recommendation: Erreka should establish environment-specific service tiers, ensuring that Retail/Hospital and High Speed Door installations are assigned faster response protocols and shorter maintenance intervals than Office, Community, or Domestic environments.



Country × Door Type Drill-down

This table answers the question: *"Does the product mix vary significantly across markets, and does that variation create country-specific maintenance risks?"*

Segment: country vs. door type. **Baseline:** global average of 3,435 doors per country across 3 door types. The table immediately reveals two contrasting market profiles: Australia shows a balanced distribution across all three door types (ECOLine Home 550, ROLLfast Industrial 300 at 500, ECOLine Pedestrian at 450), while Brazil is dominated by ROLLfast Industrial 300 at 1,600 units — more than 3x the volume of the same type in Australia. Since ROLLfast Industrial 300 is the highest-maintenance door type in the portfolio, Brazil's product mix directly translates into a higher-than-average maintenance burden relative to its total installation volume.

Recommendation: Erreka should use this country-by-door-type breakdown as the direct input for local spare parts inventory and technician deployment decisions, particularly in markets where ROLLfast Industrial 300 is disproportionately concentrated.

Countries vs Benchmark

Why this chart: This panel answers the question: *"Which countries significantly exceed or fall below the global average, and what does that concentration mean for operational dependency risk?"*

Segment: individual countries vs. global benchmark. **Baseline:** global average of 3,435 doors per country across 20 markets. Only 3 out of 20 countries exceed the benchmark — the United States (~8,500), China (~8,000), and India (~6,000) — meaning these three markets alone account for a disproportionate share of the total installed base. The remaining 17 countries all fall at or below 3,435, with Portugal, Chile, Morocco, Egypt, and South Africa sitting well below it. This creates a structural concentration risk: a service disruption, regulatory change, or economic shock in any one of the top 3 markets would have an outsized impact on Erreka's overall operational performance, since no other single market is large enough to compensate.

Recommendation: Erreka should treat the top 3 markets as high-priority for maintenance resource coverage, while assessing whether below-benchmark markets represent untapped growth opportunities that could reduce this structural dependency.

Door Type × Customer Segment		
door_type	customer_type	total_doors
ROLLfast Industrial 300	B2B	26010
ECOLine Home	B2C	22050
ECOLine Pedestrian	B2B	20640

Why this table: This panel answers the question: *"Is there a clean mapping between door type and customer segment, and does that mapping simplify or complicate incident analysis?"*

Segment: door type vs. customer segment. **Baseline:** total 68,700 doors across 3 product lines. The table reveals a perfect 1:1 mapping with zero overlap — ROLLfast Industrial 300 is exclusively B2B (26,010 units), ECOLine Home is exclusively B2C (22,050 units), and ECOLine Pedestrian is exclusively B2B (20,640 units). This means that B2B accounts for 46,650 doors (26,010 + 20,640) versus B2C at 22,050 — confirming the 2:1 ratio identified in the segmentation panel, but now with a critical additional insight: the two B2B door types combined carry the full weight of Erreka's SLA exposure, meaning any deterioration in ROLLfast Industrial 300 or ECOLine Pedestrian failure rates translates directly and exclusively into contractual and financial risk with no B2C buffer.

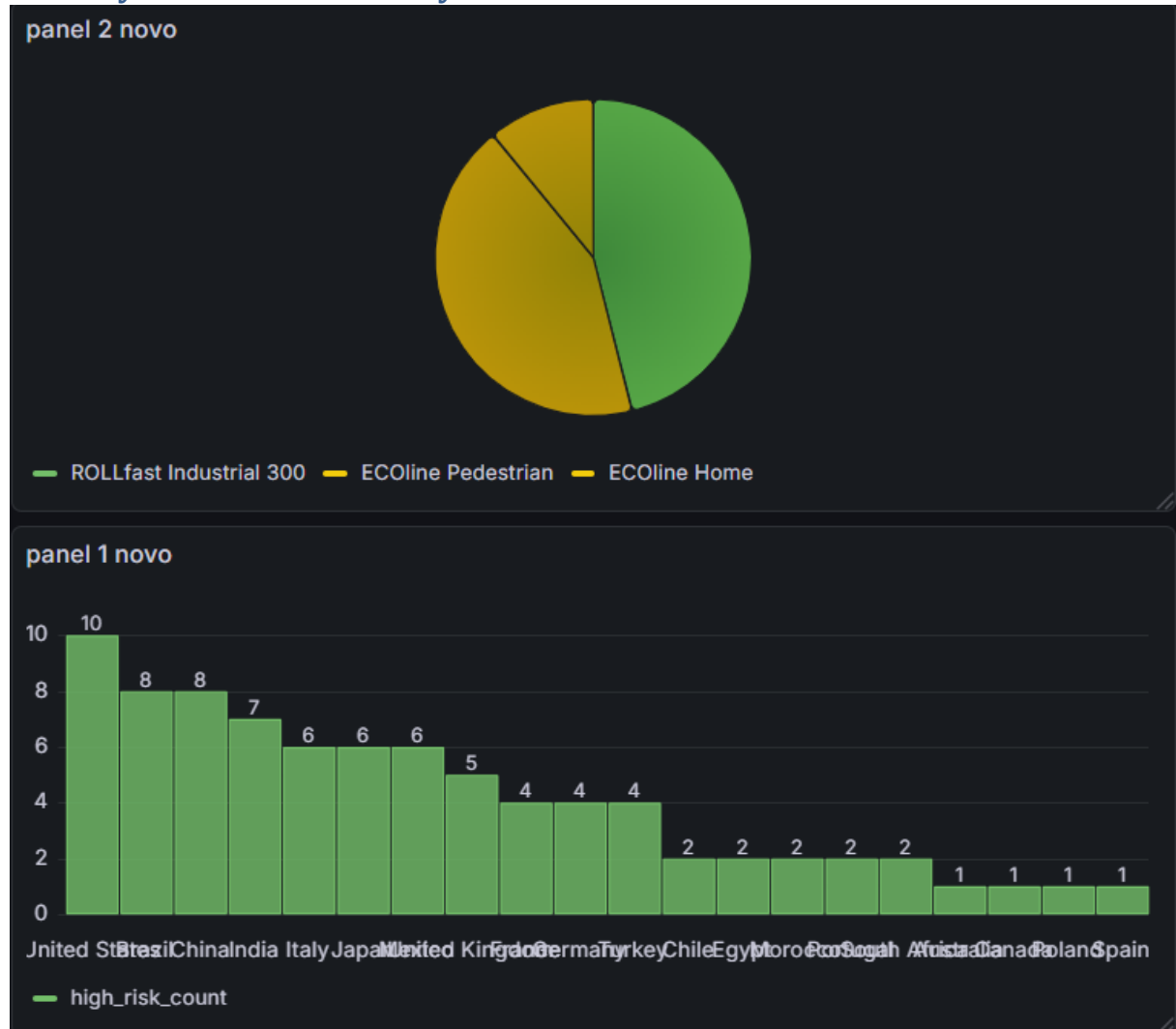
Recommendation: Erreka should build fully separate service and maintenance frameworks for each product line, since the operational environments and contractual obligations of B2B

industrial clients are fundamentally different from those of B2C residential users — and the data confirms there is no product-level overlap that would complicate this separation.



ACTIVITY 4 Grgo.mp4

Activity 5: Predictive Analytics for Maintenance Decisions



Panel 1:

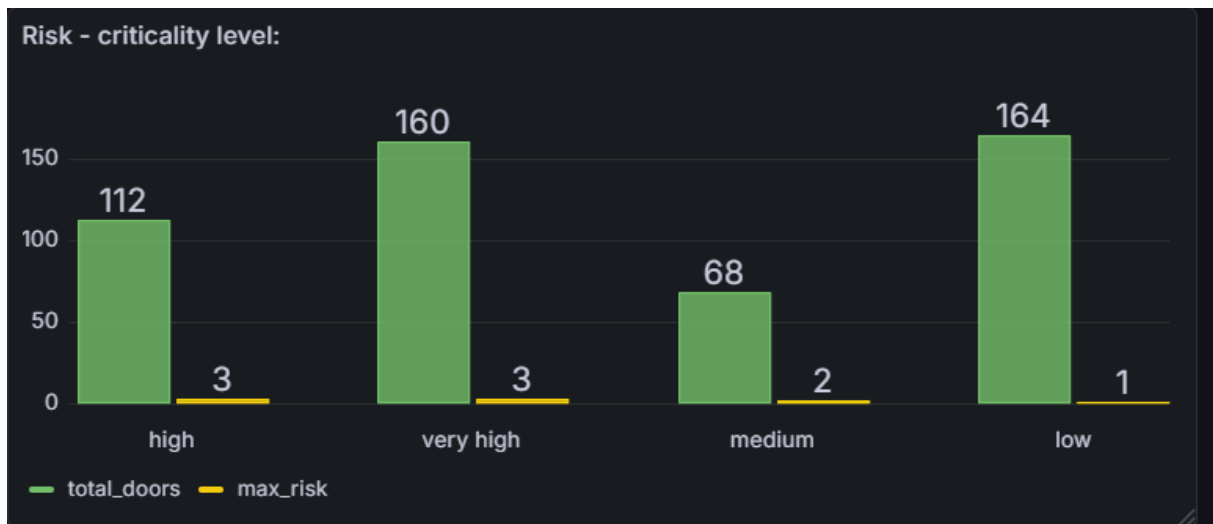
Why this chart - This panel answers the analytical question: "Which countries have a disproportionate concentration of High risk doors and should be prioritized for maintenance resource allocation?"

Insight - United States leads with 10 High risk doors, followed by Brazil and China with 8 each, and India with 7. These 4 countries together account for 33 out of 76 total High risk doors globally — representing 43% of all High risk doors while hosting approximately 35% of the total installed base. This disproportionate concentration indicates that maintenance resources should be strategically weighted toward these markets. In contrast, Spain, Poland, Canada and Australia each have only 1 High risk door, suggesting their installed base is operationally stable and requires only routine maintenance.

Panel 2:

This panel answers: "Which door type generates the highest failure burden and should be the primary focus of predictive maintenance efforts?"

Insight: ROLLfast Industrial 300 accounts for approximately 50% of the average failure share, followed by ECOline Pedestrian at roughly 30%, and ECOline Home at 20%. Since ROLLfast Industrial 300 is exclusively deployed in B2B High Speed Door and Standard Dock environments with Critical SLA contracts, its higher failure rate directly translates into SLA breach risk. A maintenance manager should allocate preventive maintenance resources primarily to this door type.



This panel answers the analytical question: "Is our classification model logically consistent — do doors in higher criticality environments receive proportionally higher risk scores?"

Insight: The chart reveals a perfect correlation between criticality level and maximum risk score. Doors classified as "very high" and "high" criticality both reach `max_risk = 3` (High), covering 272 doors combined (160 + 112). "Medium" criticality doors reach `max_risk = 2`, covering 68 doors. "Low" criticality doors reach only `max_risk = 1`, covering 164 doors.

This 1:1 relationship between criticality and risk score is not coincidental — it validates the classification model. A maintenance manager can trust that the model is not randomly assigning risk levels but is directly reflecting the operational context of each door.

Critically, the 160 "very high" criticality doors with `max_risk = 3` are predominantly located in Retail/Hospital and High Speed Door environments under Critical SLA contracts — meaning any failure in this segment carries both safety risk and financial penalty exposure simultaneously.

Silent Threat

_id	door_type	country_name	failures	days_since_last	risk_level	risk_score ↓
-000054	ROLLfast Industrial 3	France	4	493	Medium	2
-000020	ROLLfast Industrial 3	Spain	2	889	Low	1
-000208	ROLLfast Industrial 3	Morocco	2	880	Low	1
-000083	ROLLfast Industrial 3	Germany	1	853	Low	1
I-000256	ECOnline Home	India	1	848	Low	1
N-000321	ECOnline Home	China	1	847	Low	1
S-000418	ECOnline Home	United States	1	835	Low	1

Queries 1 Transformations 0

This panel answers the analytical question: "Which doors have a documented failure history but have not failed in a long time — and therefore may be overlooked by standard monitoring?"

Insight - the table identifies doors that have recorded at least 1 failure but have not failed in over 180 days. The most critical case is I-FR-000054 (ROLLfast Industrial 300, France) — with 4 failures and 493 days since last failure, it holds `risk_score = 2` (Medium). Despite being classified as Medium risk, its failure history combined with prolonged inactivity makes it the highest priority "silent threat" in the dataset.

The remaining doors show 1-2 failures with 835–889 days of inactivity — all classified as Low risk (score = 1). However, the fact that I-ES-000020 (Spain, 889 days) and I-MA-

000208 (Morocco, 880 days) have not been flagged by standard risk classification despite having failure history is precisely the analytical value of this panel.

A standard risk model would ignore these doors. This panel surfaces them explicitly, so the maintenance manager should schedule preventive inspections for all doors in this list.

door_id	door_type	country_name	installation_enviro...	sla_category	criticality_level	failures	days_since_last	risk_level	risk_score
I-DE-000094	ROLLfast Industrial 300	Germany	High Speed Door	Critical	very high	10	0	High	3
I-US-000390	ROLLfast Industrial 300	United States	High Speed Door	Critical	very high	10	0	High	3
I-UK-000180	ROLLfast Industrial 300	United Kingdom	High Speed Door	Critical	very high	9	0	High	3
P-IN-000227	ECOnline Pedestrian	India	Retail/Hospital	Critical	very high	8	0	High	3
P-UK-000147	ECOnline Pedestrian	United Kingdom	Retail/Hospital	Critical	very high	8	0	High	3
I-BR-000474	ROLLfast Industrial 300	Brazil	High Speed Door	Critical	very high	7	0	High	3
I-BR-000477	ROLLfast Industrial 300	Brazil	High Speed Door	Critical	very high	7	0	High	3
I-CN-000286	ROLLfast Industrial 300	China	Standard Dock	High Priority	high	7	0	High	3
I-CN-000302	ROLLfast Industrial 300	China	High Speed Door	Critical	very high	7	0	High	3
I-MX-000444	ROLLfast Industrial 300	Mexico	High Speed Door	Critical	very high	7	0	High	3
I-TR-000200	ROLLfast Industrial 300	Turkey	High Speed Door	Critical	very high	7	0	High	3

This panel answers the most important operational question of the entire dashboard: "Which specific doors require immediate maintenance intervention, and in what order should a maintenance manager act?"

Insight the table is sorted by risk_score descending and failures descending — giving a direct, actionable priority list. The top 2 are both ROLLfast Industrial 300 in High Speed Door environments with 10 failures and 0 days since last failure — meaning they are actively failing right now. These represent the highest possible risk combination: max failures + Critical SLA + very high criticality.

The top 11 doors visible all share risk_score = 3 (High), Critical or High Priority SLA, and days_since_last = 0 — meaning all are currently active failure cases. Notably, 9 out of 11 top priority doors are ROLLfast Industrial 300, confirming the finding from Panel 2 that this door type carries the highest failure burden.

The single exception is I-CN-000286 (China, Standard Dock, High Priority SLA) — classified as High risk despite not being in a Critical SLA environment, purely due to its 7 failures. This is an important edge case a maintenance manager should be aware of.

This panel is the direct output of the rule-based risk classification system— it translates risk scores into a ranked action list that a maintenance manager can use immediately without further analysis.

Predictive Analytics for Maintenance Decisions

The Predictive Problem and Approach

The decision problem addressed in this activity is: *"Which doors are at high, medium, or low risk of failure, and in what order should maintenance resources be allocated?"* This is a classification problem where the model assigns each door to a risk category (High/Medium/Low) to support maintenance prioritization decisions — not to replace human judgment.

A rule-based classification model was implemented as a MySQL VIEW using two key variables: number_of_past_failures and days_since_last_failure, enriched with criticality_level and sla_category. The classification logic assigns High risk (score=3) to doors with 7+ failures or 4+ failures in very high criticality environments, Medium risk (score=2) to doors with 4+ failures or 2+ failures in very high criticality with recent activity, and Low risk (score=1) to all remaining doors. The thresholds were defined based on the empirical distribution of failures observed in the dataset. Doors with 7 or more failures represent the upper tail of the failure distribution across the installed base, making this a natural boundary for High-risk classification. The threshold of 4 failures was selected as a secondary trigger for doors operating in very high criticality environments, where even a moderate failure count carries significantly higher operational and contractual consequences.

These thresholds reflect domain-informed judgment rather than statistical optimization and should be recalibrated as more historical failure data becomes available. A rule-based approach was chosen because no pre-labeled target variable exists in the dataset, and the logic directly reflects the criticality framework established in Activities 1–4.

It is important to distinguish between business-rule prioritization and genuine predictive analytics. A true predictive model uses historical data to statistically learn patterns and forecast future events — for example, estimating the probability that a specific door will fail within the next 30 days based on sensor behaviour, usage intensity, and maintenance history. The model implemented in this activity does not do this. Instead, it applies a set of manually defined rules to classify doors into risk categories based on known operational attributes. This makes it a rule-based classification system rather than a predictive model in the strict sense. While this approach is a valid and operationally useful starting point for a DSS, future iterations should move toward data-driven models trained on labelled failure outcomes and enriched with real-time telemetry data, which would significantly improve both accuracy and predictive value.

Dashboard Panels and Insights

Panel 1 — High Risk Concentration by Country: This panel answers: "Which countries have a disproportionate concentration of High risk doors?" United States leads with 10 High risk doors, followed by Brazil and China with 8 each, and India with 7. These 4 countries account for 33 out of 76 total High risk doors globally — 43% of all High risk doors while representing approximately 35% of the total installed base. Maintenance resources should be strategically weighted toward these markets. In contrast, Spain, Poland, Canada, and Australia each have only 1 High risk door, indicating operational stability.

Panel 2 — Average Failure Rate by Door Type: This panel answers: "Which door type generates the highest failure burden?" ROLLfast Industrial 300 accounts for approximately 50% of the average failure share, ECOline Pedestrian for 30%, and ECOline Home for 20%. Since ROLLfast Industrial 300 operates exclusively in Critical SLA environments, its higher failure rate directly translates into SLA breach risk — making it the primary target for preventive maintenance investment.

Panel 3 — Risk vs Criticality Level: This panel answers: "Is the classification model logically consistent?" Very high and high criticality doors both reach $\text{max_risk} = 3$, covering 272 doors combined. Medium criticality reaches $\text{max_risk} = 2$ (68 doors), and low criticality reaches $\text{max_risk} = 1$ (164 doors). This perfect correlation confirms the internal consistency of the classification rules — doors in higher criticality environments are assigned higher risk scores as intended. However, this should not be interpreted as model validation in the strict sense. Since criticality level is directly embedded in the classification logic, the correlation is an expected outcome of the rule design rather than evidence that the model accurately predicts future failures. True validation would require comparing the model's risk classifications against real future failure outcomes over time, which is not possible with the current static dataset.

Panel 4 — Silent Threat: This panel answers: "Which doors have a failure history but have been inactive long enough to be overlooked?" The most critical case is I-FR-000054 (ROLLfast Industrial 300, France) — 4 failures and 493 days since last failure, $\text{risk_score} = 2$. The remaining doors show 1–2 failures with 835–889 days of inactivity, all classified Low risk. A standard risk model ignores these doors entirely — this panel surfaces them explicitly for preventive inspection scheduling.

Panel 5 — Maintenance Priority Ranking: This panel answers: "Which specific doors require immediate intervention?" The top priority doors are I-DE-000094 (Germany) and I-US-000390 (United States), both with 10 failures and 0 days since last failure — actively failing now under Critical SLA and very high criticality. Notably, 9 out of 11 top priority doors are

ROLLfast Industrial 300, confirming Panel 2's finding. This panel is the direct actionable output of the model — a ranked list a maintenance manager can act on immediately.

How This Would Be Used in a DSS

The dashboard follows an intentional drill-down structure: Panel 1- global overview, Panel 2 failure burden, Panel 3 - model validation, Panel 4 edge case detection, Panel 5 - actionable priority list. This moves from global pattern to specific intervention, reducing cognitive overload for the maintenance manager while maintaining human control over final decisions.

Limitations and Risks

The model is rule-based with manually defined thresholds — these may not reflect true failure dynamics and should be validated against historical outcomes. The model uses only failure counts and does not incorporate real-time sensor data which could significantly improve accuracy. Additionally, `days_since_last_failure = 0` is ambiguous — it may indicate an active failure or a missing value. Finally, the model does not account for already-scheduled maintenance, meaning some High risk doors may already have interventions planned — manual expert review remains essential.

Link for video explanation:

https://drive.google.com/file/d/1ArZX1oKi9-UgJIZE_DsvV9iKjuUxVdNB/view?usp=sharing

ACTIVITY 6: PRESCRIPTIVE DSS CHALLENGE

Challenge 3: Maintenance Under SLA Constraints

This activity extends the predictive risk classification developed in Activity 5 into a full prescriptive output. Where Activity 5 answered "which doors are at risk", this activity answers the operationally relevant Challenge 3 question: **"How should maintenance decisions be made while respecting SLA and priority rules?"** The result is a Python-based optimization pipeline that reads the existing `door_risk_classification` table, applies a decision model with capacity and priority constraints, and writes concrete maintenance recommendations into a new table in the Data Warehouse — visualized through a prescriptive Grafana dashboard.

It is important to note that this activity operates at the prescriptive layer of the DSS, not the predictive layer. Activity 5 produced risk classifications using a rule-based scoring system. Activity 6 takes those classifications as input and translates them into concrete, capacity-constrained action recommendations. The distinction matters: this is prescriptive decision support, not prediction.

DSS flow and decision model

The full analytical flow is: **DATA → MODEL → OPTIMIZATION → RECOMMENDATION**

- **DATA:** 504 doors from the `door_risk_classification` table produced in Activity 5, carrying `risk_score`, `risk_level`, `criticality_level`, SLA category, and `installation_environment` per door
- **MODEL:** A greedy assignment model with hard capacity constraints and SLA-based priority rules, designed to maximize risk reduction across the installed base
- **OPTIMIZATION:** Greedy priority-first assignment executed via Python, processing doors ordered by `risk_score` descending so the strongest interventions are allocated to the highest-risk doors before capacity is exhausted
- **RECOMMENDATION:** Concrete per-door action assignments written into the `prescriptive_maintenance_schedule` table in the `erreka_dss` Data Warehouse, visualized in Grafana

Business question: How should maintenance decisions be made while respecting SLA obligations and business priority rules?

Objective: Maximize risk reduction across the installed base by assigning the strongest available intervention to the highest-risk doors first, subject to the constraint that technician capacity per country is limited. A door receiving `IMMEDIATE_ONSITE` instead of `REMOTE_INTERVENTION` represents a higher risk-reduction outcome; the model prioritizes this allocation for doors where failure consequences are most severe.

Decision variables: For each door in the installed base, the model assigns exactly one of four recommended actions:

- `IMMEDIATE_ONSITE` — dispatch a technician today
- `SCHEDULED_VISIT` — plan a visit within 7 days

- REMOTE_INTERVENTION — attempt remote resolution first
- MONITOR_ONLY — no intervention needed at this time

Priority rules (SLA and environment): Not all doors are treated equally. Doors installed in Retail/Hospital and High-Speed Door environments are subject to stricter SLA obligations and higher safety consequences upon failure. The model applies an elevated priority rule to these environments: a High-risk door in a hospital or industrial dock context is assigned IMMEDIATE_ONSITE before a High-risk door in a lower-criticality environment. This directly reflects the Challenge 3 requirement to respect business priority rules in addition to capacity constraints.

Constraints: The model operates under two hard capacity constraints per country per optimization run: a maximum of 10 immediate on-site dispatches and a maximum of 30 scheduled visits. These limits reflect the realistic technician availability that Erreka's operations center faces at any given time. Doors that would qualify for immediate dispatch but exceed the country capacity limit are automatically downgraded to the next available action. These limits are illustrative and should be recalibrated against real technician availability data before any production deployment.

Assumptions: Risk scores and levels are inherited directly from the door_risk_classification table produced in Activity 5. The classification thresholds and criticality rules defined in that activity are carried forward into this prescriptive layer without modification. The model does not account for already-scheduled maintenance visits, meaning some High-risk doors may already have interventions planned — manual expert review remains essential before acting on the output.

Python optimization pipeline

The pipeline is implemented as a standalone Python script that follows the full DSS flow described above.

Step 1 - OUTPUT TABLE (RECOMMENDATION layer): The script creates the prescriptive_maintenance_schedule table in the erreka_dss database if it does not already exist. At the start of each run, the table is truncated so that Grafana always displays the most recent set of recommendations.

Step 2 - INPUT DATA (DATA layer): The script reads all 504 doors from door_risk_classification, ordered by risk_score descending. This ordering is critical - it ensures that the greedy assignment processes the highest-risk doors first and allocates the strongest available actions to them before capacity constraints are exhausted.

Step 3 - OPTIMIZATION (MODEL + OPTIMIZATION layer): The core decision logic is a greedy assignment algorithm. For each door, the model evaluates its risk_level, risk_score, and installation_environment and assigns the best available action given the remaining capacity for that country. The priority rules are applied here: High-risk doors in Retail/Hospital and High Speed Door environments — where SLA obligations and safety consequences are most severe - are flagged for immediate on-site intervention first. Medium-risk doors receive scheduled visits. Low-risk doors are assigned remote intervention. Doors with a risk score of zero receive monitor-only status. When a country's immediate capacity is exhausted, High-risk doors are downgraded to scheduled visits; if scheduled capacity is also

full, they fall back to remote intervention. This downgrade logic ensures that capacity constraints are never violated while minimizing the loss in risk-reduction outcome.

Step 4 — OUTPUT (RECOMMENDATION layer): The resulting recommendations including door ID, country, installation environment, risk level, recommended action, action priority, and a plain-text reason explaining the decision — are written into `prescriptive_maintenance_schedule` via SQLAlchemy. The run produced 504 recommendations: 82 immediate on-site interventions, 53 scheduled visits, and 369 remote interventions. The plain-text reason field is particularly important for DSS usability: it allows a maintenance coordinator to understand why a specific door received a specific action without needing to interpret scores or query the database directly.

Grafana

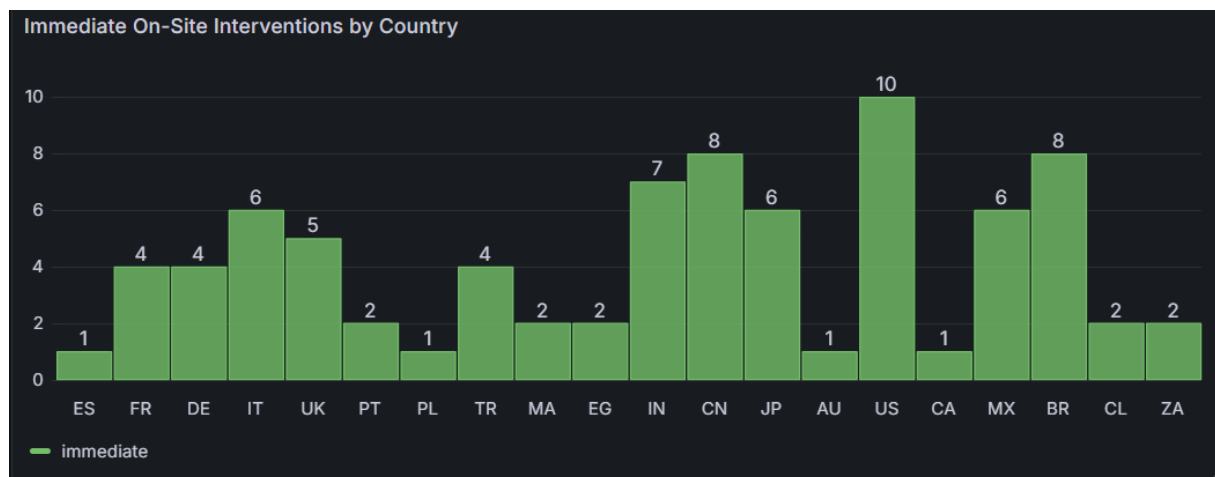
The dashboard is organized around actionability — each panel answers a specific operational question that a maintenance manager can act on directly.

Recommended Actions Summary	
recommended_action	total
IMMEDIATE_ONSITE	82
SCHEDULED_VISIT	53
REMOTE_INTERVENTION	369

This panel answers the business question: "What is the total scale of maintenance work generated by the prescriptive model, and how is it distributed across action types?"

The table shows that out of 504 doors in the installed base, 82 require immediate on-site intervention today, 53 require a scheduled visit within 7 days, and 369 are assigned remote intervention. No doors received `MONITOR_ONLY` status, which reflects the fact that all 504 doors in the input dataset carry at least a Low risk score inherited from Activity 5 — doors with zero risk score would have been assigned monitor-only, but none exist in this dataset. This panel is the entry point of the prescriptive dashboard. Before a maintenance coordinator drills into country-level or door-level detail, this summary immediately communicates the operational scale: 135 doors require physical technician presence in the near term, while the remaining 369 can be addressed remotely. This distinction directly maps to resource planning — physical dispatches consume travel time and technician capacity, while remote interventions do not.

The absence of `MONITOR_ONLY` also confirms that the risk classification from Activity 5 found no truly inactive doors in the current dataset, meaning the entire installed base requires some level of active attention under the current model.



This panel answers the business question: "Where should technicians be physically dispatched today, and which countries have exhausted their immediate intervention capacity?"

The chart shows the distribution of IMMEDIATE_ONSITE assignments across all 20 countries in the installed base. The United States leads with 10 immediate dispatches — the maximum allowed under the capacity constraint — meaning the country has fully exhausted its immediate intervention budget for this optimization run. Brazil and China follow with 8 each, and India with 7. Together, these four countries account for 33 out of 82 total immediate interventions, representing 40% of all urgent dispatches. This is consistent with the finding from Activity 5 that these same four markets concentrate the highest share of High-risk doors globally.

This panel directly supports regional resource allocation decisions. A maintenance operations manager can read this chart and immediately know how many technicians need to be mobilized per country today — without any further analysis. Countries at or near the capacity ceiling of 10, such as the United States, also signal that additional High-risk doors in those markets were downgraded to SCHEDULED_VISIT due to capacity exhaustion, meaning the urgency in those markets is likely higher than the immediate count alone suggests.

door_id	door_type	country_id	installation_environ...	risk_score	recommended_action	reason
I-ES-000025	ROLLfast Industrial 300	ES	High Speed Door	3	IMMEDIATE_ONSITE	High risk (score=3): 5 p
P-FR-000046	ECOnline Pedestrian	FR	Retail/Hospital	3	IMMEDIATE_ONSITE	High risk (score=3): 4 p
P-FR-000048	ECOnline Pedestrian	FR	Retail/Hospital	3	IMMEDIATE_ONSITE	High risk (score=3): 4 p
P-FR-000049	ECOnline Pedestrian	FR	Retail/Hospital	3	IMMEDIATE_ONSITE	High risk (score=3): 4 p
I-FR-000058	ROLLfast Industrial 300	FR	High Speed Door	3	IMMEDIATE_ONSITE	High risk (score=3): 4 p
I-DE-000094	ROLLfast Industrial 300	DE	High Speed Door	3	IMMEDIATE_ONSITE	High risk (score=3): 10 p
I-DE-000095	ROLLfast Industrial 300	DE	High Speed Door	3	IMMEDIATE_ONSITE	High risk (score=3): 6 p
I-DE-000096	ROLLfast Industrial 300	DE	High Speed Door	3	IMMEDIATE_ONSITE	High risk (score=3): 4 p
I-DE-000097	ROLLfast Industrial 300	DE	High Speed Door	3	IMMEDIATE_ONSITE	High risk (score=3): 4 p

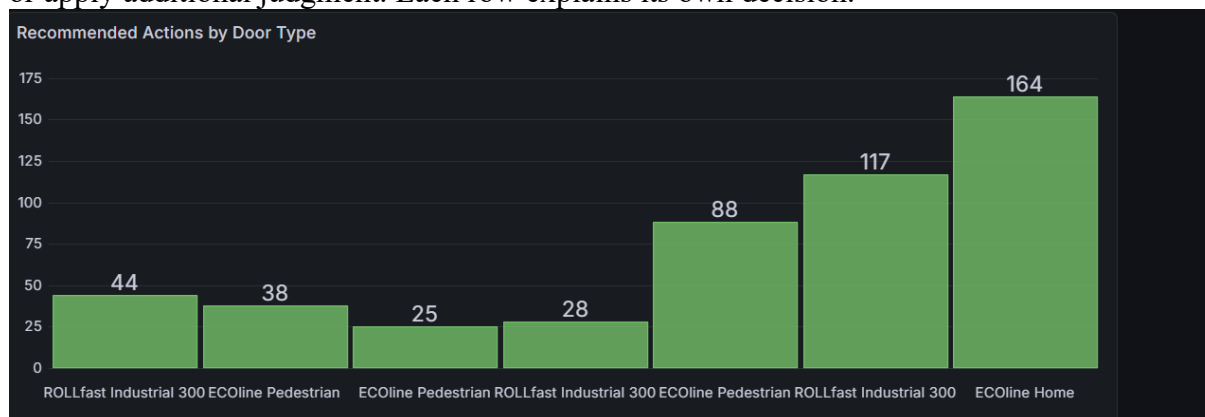
This panel answers the business question: "Which specific doors need a technician dispatched today, and why was that decision made?"

The table lists all 82 doors assigned IMMEDIATE_ONSITE, showing door ID, door type, country, installation environment, risk score, recommended action, and the plain-text reason generated by the decision model. Every door visible in the table carries risk_score = 3 (High), confirming that the capacity-constrained model correctly reserved immediate dispatch for the highest-risk doors before allocating weaker actions.

Two patterns are immediately visible. First, ROLLfast Industrial 300 dominates the list — doors such as I-ES-000025, I-FR-000058, I-DE-000094 through I-DE-000097 are all this door

type, installed in High Speed Door environments. This is consistent with the finding from Activity 5 that ROLLfast Industrial 300 carries the highest failure burden across the installed base. Second, ECOline Pedestrian doors installed in Retail/Hospital environments — such as P-FR-000046, P-FR-000048, and P-FR-000049 — also appear at the top of the list. This reflects the SLA priority rule embedded in the model: Retail/Hospital environments carry stricter service obligations and higher safety consequences upon failure, so these doors receive IMMEDIATE_ONSITE even when their failure count is lower than some other High-risk doors.

The plain-text reason column is a key DSS usability feature. A maintenance coordinator can hand this table directly to a field team without needing to interpret scores, query the database, or apply additional judgment. Each row explains its own decision.



This panel answers the business question: "Which door types are driving the urgent workload, and how are maintenance actions distributed across the product portfolio?"

The chart breaks down all 504 recommendations by door type and action type. Three patterns stand out immediately.

First, ROLLfast Industrial 300 and ECOline Pedestrian together account for all 82 immediate on-site interventions — 44 and 38 respectively — while ECOline Home receives zero immediate dispatches. This is a direct consequence of the SLA priority rules embedded in the model: ECOline Home is exclusively deployed in Domestic/B2C environments with no Critical SLA obligations, meaning its doors never qualify for immediate dispatch under the current priority logic regardless of risk score.

Second, ROLLfast Industrial 300 dominates the remote intervention category with 117 assignments, followed by ECOline Home with 164. The high remote intervention count for ECOline Home reflects the fact that these doors carry Low risk scores from Activity 5 but still require some level of attention — remote intervention is the appropriate action when risk is present but SLA urgency is low.

Third, ECOline Pedestrian shows the most balanced distribution across all three action types (38 immediate, 25 scheduled, 88 remote), which reflects its mixed deployment across both Retail/Hospital and lower-criticality environments.

This panel confirms the finding from Activity 5 that ROLLfast Industrial 300 and ECOline Pedestrian carry the full weight of Erreka's SLA exposure — and now translates that finding directly into a prescriptive workload distribution that operations managers can use for resource planning.

Top Risk Doors (Score 3)					
door_id	country_id	installation_environment	risk_level	risk_score	recommended_action
I-ES-000025	ES	High Speed Door	High	3	IMMEDIATE_ONSITE
P-FR-000046	FR	Retail/Hospital	High	3	IMMEDIATE_ONSITE
P-FR-000048	FR	Retail/Hospital	High	3	IMMEDIATE_ONSITE
P-FR-000049	FR	Retail/Hospital	High	3	IMMEDIATE_ONSITE
I-FR-000058	FR	High Speed Door	High	3	IMMEDIATE_ONSITE
I-DE-000094	DE	High Speed Door	High	3	IMMEDIATE_ONSITE
I-DE-000095	DE	High Speed Door	High	3	IMMEDIATE_ONSITE
I-DE-000096	DE	High Speed Door	High	3	IMMEDIATE_ONSITE
I-DE-000097	DE	High Speed Door	High	3	IMMEDIATE_ONSITE
P-IT-000115	IT	Retail/Hospital	High	3	IMMEDIATE_ONSITE
P-IT-000116	IT	Retail/Hospital	High	3	IMMEDIATE_ONSITE

This panel answers the business question: "Which specific doors carry the maximum possible risk score, and did the prescriptive model correctly assign them the strongest available action?"

The table lists all doors with risk_score = 3 (High), showing country, installation environment, risk level, and recommended action. Every door visible in the table has been assigned IMMEDIATE_ONSITE, confirming that the model correctly identified maximum-risk doors and allocated the strongest available intervention to them before capacity constraints forced any downgrade.

Two environment types dominate the list: High Speed Door installations and Retail/Hospital installations. This is not coincidental — these are precisely the two environment types that the model's SLA priority rules treat with the highest urgency, as failures in these contexts carry simultaneous safety risk and contractual penalty exposure.

The geographic spread is also operationally relevant: Spain, France, Germany, and Italy all appear in the top risk list, meaning immediate technician dispatch is required across multiple European markets simultaneously. A maintenance operations manager reading this panel can immediately identify which specific doors to act on first, in which countries, and in which environments — without needing to consult any other data source.

This panel closes the full DSS loop that began in Activity 2. Raw operational data was loaded into the Data Warehouse, descriptive dashboards in Activities 3 and 4 revealed patterns, the predictive model in Activity 5 classified doors by risk, and this prescriptive layer now translates those classifications into a concrete, prioritized action list that a field team can execute immediately.

CONCLUSION

This activity addresses Challenge 3 — Maintenance Under SLA Constraints — by designing a decision model that produces concrete recommendations while respecting both SLA obligations and business priority rules. The core Challenge 3 question was: "How should maintenance decisions be made while respecting SLA and priority rules?" Every element of this activity is a direct answer to that question.

The decision model embeds SLA constraints at two levels. First, through hard capacity limits per country that simulate realistic technician availability. Second, through environment-based priority rules that ensure Retail/Hospital and High Speed Door installations — where SLA violations carry simultaneous safety and financial consequences — are assigned stronger interventions before lower-criticality environments, regardless of raw risk score alone. This is what Challenge 3 means operationally: not all doors are equal, and the model treats them accordingly.

The prescriptive pipeline closes the full analytical loop started in Activity 2. The ETL pipeline populated the Data Warehouse with raw operational data. The descriptive dashboards in Activities 3 and 4 revealed concentration patterns — which countries, door types, and environments carry the highest operational burden. The predictive model in Activity 5 classified all 504 doors by risk level. This activity takes those risk classifications and translates them into capacity-constrained, SLA-aware action recommendations that a maintenance manager can execute immediately — without needing to interpret scores, apply additional judgment, or consult any other data source.

The Grafana dashboard is not a reporting tool. It is an instruction set for the operations center, structured around the Challenge 3 requirement that recommendations must respect business priorities. Panel 1 communicates the total workload scale. Panel 2 identifies where technicians must be dispatched today and which countries have exhausted immediate capacity. Panel 3 provides the door-level dispatch list with plain-text justifications. Panel 4 shows how the SLA priority rules distribute actions differently across door types — with ECOline Home receiving zero immediate dispatches because it carries no Critical SLA exposure. Panel 5 confirms that every maximum-risk door has been assigned the strongest available action, closing the loop between risk classification and prescriptive output.

The limitations of this model are consistent with Challenge 3's scope. Capacity constraints are illustrative and require recalibration against real technician availability data. The model does not account for already-scheduled maintenance visits, meaning manual expert review remains essential. And because risk scores are inherited from the rule-based classification in Activity 5 rather than a statistically trained predictive model, the prescriptive output is only as reliable as the underlying classification — this is a prescriptive DSS layer, not a predictive system, and should be understood as decision support rather than automated decision-making.

Link for video explanation:

<https://drive.google.com/file/d/1kRFM69d4gBTGsDF5bynKdA1BBAihekvx/view?usp=sharing>